Sri Lanka Institute of Information Technology

Application Frameworks

Assignment 01

Group ID – Y3S1-SE-104

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Testing Instruction Report

1. Unit Tests Setup and Execution

Unit tests verify the functionality of individual components or services in isolation.

Commands:

1. Run Unit Tests for a Specific File

npx jest <relative-path-to-test-file>

2. Run All Unit Tests

npx jest

3. Run Tests with Coverage

npx jest –coverage

2. Integration Testing Setup and Execution

Integration tests ensure that different modules or services work together as expected.

Setup

1. Database Configuration

- Use a test database for integration tests.
- Update the .env.test file:

```
NODE_ENV=test
DB_HOST=localhost
DB_USER=test_user
DB_PASS=test_password
DB_NAME=test_database
```

2. Seed Test Data

Populate the test database with sample data:

npm run seed

3. Mock External Services

Use libraries like:

- nock
- jest-mock to mock external API calls.

4. Commands:

1. Run Integration Tests for a Specific File

npx jest <relative-path-to-integration-test-file>

2. Run All Integration Tests

npx jest -- testPathPattern=integration

3. Performance Testing Setup and Execution

Performance tests evaluate the speed, scalability, and stability of the application under load.

Setup

1. **Install Artillery:**

   npm install -g artillery

2. **Create a Performance Test Script, run and Review the output for response times, error rates, and throughput.**

Commands:

1. **Run Performance Tests**

   artillery run performance-test.yml

4. Testing Environment Configuration

The testing environment ensures consistency and reliability during test execution.

1. **Environment Variables**

   Use a .env.test file for testing:

   NODE_ENV=test
   DB_HOST=localhost
   DB_USER=test_user
   DB_PASS=test_password
   DB_NAME=test_database

2. **Dependencies**

   Ensure all dependencies are installed:

   npm install

3. **Mocking and Stubbing**

   Use libraries like:

   - jest-mock
   - sinon

   to mock dependencies and isolate tests.

4. **Test Database**

   - Use a separate database for testing
   - Ensure it does not affect production data

5. **Reset State Before Each Test**

   Reset the database before each test to ensure isolation:

   beforeEach(async () => {

```
  await resetDatabase();
});
```

5. File-Specific Testing Instructions

These examples represent a typical project structure and can be adapted to any module in the system.

1. Service Layer Unit Tests (Example: featureService.test.js)

Purpose

Unit tests for service layer business logic functions.

Example File

src/services/featureService.test.js

Commands

01. Run

npx jest src/services/featureService.test.js

02. Run with Coverage

npx jest src/services/featureService.test.js --coverage

2. Route Layer Unit Tests (Example: featureRoute.test.js)

Purpose

Unit tests for API endpoints and route handlers.

Example File

src/routes/featureRoute.test.js

Commands

01. Run

npx jest src/routes/featureRoute.test.js

02. Run with Coverage

npx jest src/routes/featureRoute.test.js --coverage

3. Integration Tests (Example: feature.integration.test.js)

Purpose

Integration tests for verifying interaction between services, routes, and database.

Example File

src/tests/integration/feature.integration.test.js

Commands

01. Run Single Integration Test

npx jest src/tests/integration/feature.integration.test.js

02. Run All Integration Tests:

npx jest --testPathPattern=integration

4. Performance Testing

Purpose

Evaluate performance of API endpoints under load.

Example Target

/api/feature

Commands

01. Run Performance Test:

artillery run performance-test.yml

1. Testing Report
    1. Unit Testing
        1. Application & matching

```
PASS src/tests/unit/applicationController.unit.test.js
  applicationController — getMyApplications
    √ returns 200 with applications array (2 ms)
    √ returns 500 on db error (28 ms)
  applicationController — checkApplicationStatus
    √ returns hasApplied: true when application found
    √ returns hasApplied: false when application not found
    √ returns 500 on error (4 ms)
  applicationController — getApplicationById
    √ returns 404 when application not found (1 ms)
    √ returns 200 for the owner
    √ returns 403 for non-owner non-privileged user
    √ admin can view any application
  applicationController — withdrawApplication
    √ returns 404 if application not found
    √ returns 403 if user does not own the application
    √ returns 400 if application status is not "Applied"
    √ returns 200 and deletes application successfully (1 ms)
  applicationController — getApplicationsByInternship
    √ returns 200 with applications list
    √ returns 500 on error

Test Suites: 1 passed, 1 total
Tests:       15 passed, 15 total
Snapshots:   0 total
Time:        0.749 s, estimated 1 s
Ran all test suites matching /applicationController.unit/i.
```

        2. Auth

```
PASS src/tests/unit/authController.unit.test.js
  authController — register
    √ returns 400 if required fields are missing (3 ms)
    √ returns 409 if email already exists (1 ms)
    √ returns 400 if role is invalid
    √ returns 201 and token on successful registration (1 ms)
    √ returns 201 with valid role "organization"
    √ returns 500 on db error (21 ms)
  authController — login
    √ returns 400 if email or password missing
    √ returns 401 if user not found
    √ returns 401 if password does not match (1 ms)
    √ returns 200 and token on successful login
    √ returns 500 on unexpected error (2 ms)
  authController — getMe
    √ returns 401 if req.user is missing (1 ms)
    √ returns 200 with sanitized user data
  authController — updateMe
    √ returns 401 if not authenticated
    √ returns 400 if nothing to update
    √ returns 200 after updating email
    √ returns 200 after updating password
  authController — deleteUser
    √ returns 401 if not authenticated
    √ returns 403 if non-admin tries to delete another user
    √ allows user to delete own account (1 ms)
    √ allows admin to delete any account

Test Suites: 1 passed, 1 total
Tests:       21 passed, 21 total
Snapshots:   0 total
Time:        0.559 s, estimated 1 s
Ran all test suites matching /authController.unit/i.
```

3. Profiles & Governance

```
PASS tests/unit/youthProfile.controller.test.js
  Youth profile controller (profileController)
    createProfile
      √ returns 201 and profile when creation succeeds (4 ms)
      √ returns 400 when profile already exists
      √ returns 500 when database throws (30 ms)
    getProfileByUserId
      √ returns 200 with full profile for owner (1 ms)
      √ returns 404 when profile is missing
      √ returns 200 with summary for organization viewing another user
      √ returns 500 when query fails (5 ms)
    getProfiles
      √ returns 200 with all profiles for admin
      √ returns 200 with projected profiles for organization (1 ms)
      √ returns 200 with own profile query for youth (1 ms)
      √ returns 500 when list query throws (10 ms)
    updateProfile
      √ returns 200 when owner updates successfully (2 ms)
      √ returns 404 when profile does not exist (1 ms)
      √ returns 500 when save fails (11 ms)
    uploadCv
      √ returns 200 when document metadata is valid (1 ms)
      √ returns 400 when validateDocumentMetadata rejects payload (1 ms)
      √ returns 404 when profile is missing (1 ms)
      √ returns 500 when save throws (5 ms)
    deleteProfile
      √ returns 200 when delete succeeds (1 ms)
      √ returns 500 when delete throws (5 ms)
Test Suites: 1 passed, 1 total
Tests:       20 passed, 20 total
Snapshots:   0 total
Time:        0.753 s
Ran all test suites matching /tests\\unit\\youthProfile.controller.test.js/i.
```

```
PASS tests/unit/orgProfile.controller.test.js
  Organization profile controller (organizationController)
    createOrganizationProfile
      √ returns 201 with organization when create succeeds (3 ms)
      √ returns 400 when profile already exists (1 ms)
      √ returns 500 when persistence fails (1 ms)
    getOrganizationProfileById
      √ returns 200 with organization when found (1 ms)
      √ returns 404 when organization is missing (1 ms)
      √ returns 500 when query throws
    getOrganizationProfiles
      √ returns 200 with all organizations for admin
      √ returns 200 with scoped query for organization role
      √ returns 403 for disallowed roles (1 ms)
      √ returns 500 when list query fails
    updateOrganizationProfile
      √ returns 200 when owner updates allowed fields
      √ returns 200 when admin sets verified flag
      √ does not change verified when non-admin sends verified in body (1 ms)
      √ returns 404 when organization id does not exist
      √ returns 403 when another user is not admin (1 ms)
      √ returns 500 when save fails
    uploadOrganizationDocument
      √ returns 200 when document metadata is valid (1 ms)
      √ returns 400 when validateOrgDocumentMetadata rejects
      √ returns 404 when organization is missing
      √ returns 403 when user is not owner or admin
      √ returns 500 when save throws (1 ms)
    deleteOrganizationProfile
      √ returns 200 when owner deletes successfully
      √ returns 404 when organization does not exist
      √ returns 403 when caller is not owner or admin
      √ returns 500 when delete throws

Test Suites: 1 passed, 1 total
Tests:       25 passed, 25 total
Snapshots:   0 total
Time:        0.656 s, estimated 1 s
Ran all test suites matching /tests\\unit\\orgProfile.controller.test.js/i.
```

4. Internship

```
PS C:\Users\aathi\OneDrive\Documents\Aathika\My_Projects\SDG-GOAL---Decent-work-and-economic-growth\backend> npx
 jest src/services/internshipService.test.js
    √ should not attach coordinates when geocode returns null (1 ms)
    √ should not send email when org has no email
  updateInternship
    √ should return null if internship not found
    √ should send posted email when transitioning Draft → Active
    √ should send closed email when transitioning Active → Closed
    √ should send updated email when non-status fields change on Active internship
    √ should geocode location on update (1 ms)
  deleteInternship
    √ should delete internship by id and organizationId (1 ms)
    √ should return null if internship does not exist
  getMyInternshipsService
    √ should return all internships for an organization
    √ should filter by status when provided (1 ms)
  incrementViewCountService
    √ should increment view count and return updated internship
    √ should throw if internship not found (1 ms)
  getDashboardStatsService
    √ should return aggregated dashboard stats (1 ms)
    √ should return acceptanceRate of 0 when no applicants (1 ms)
  searchInternshipsService
    √ should return paginated internships
    √ should build $or filter when keyword provided
    √ should apply skills filter when skills param provided (1 ms)
    √ should use $geoWithin filter when location geocodes successfully
    √ should fall back to regex location filter when geocode fails (1 ms)

Test Suites: 1 passed, 1 total
Tests:       27 passed, 27 total
Snapshots:   0 total
Time:        1.45 s, estimated 2 s
```

```
 PASS  src/tests/unit/internshipController.unit.test.js
  internshipController — createInternshipController
    √ returns 201 with created internship (2 ms)
    √ returns 500 on service error (1 ms)
  internshipController — updateInternshipController
    √ returns 200 with updated internship (1 ms)
    √ returns 500 on service error (1 ms)
  internshipController — deleteInternshipController
    √ returns 200 with success message
    √ returns 500 on error (1 ms)
  internshipController — getInternshipByIdController
    √ returns internship correctly
    √ returns 500 if not found (1 ms)
  internshipController — getMyInternships
    √ passes status query param to service
    √ returns 500 on error
  internshipController — incrementViewCountController
    √ returns success with updated internship
    √ returns 500 on error
  internshipController — getDashboardStats
    √ returns stats correctly
    √ returns 500 on error (1 ms)
  internshipController — searchInternshipsController
    √ returns search results
    √ returns 500 on service error (1 ms)
  internshipController — getNearbyInternshipsController
    √ returns empty + locationAvailable:false if profile missing coordinates (1 ms)
    √ returns nearby internships when profile has coordinates
    √ returns 500 on error (26 ms)

Test Suites: 1 passed, 1 total
Tests:       19 passed, 19 total
Snapshots:   0 total
Time:        0.536 s, estimated 1 s
Ran all test suites matching /internshipController.unit/i.
```

5. Course

```
PASS  src/tests/unit/courseController.unit.test.js
  courseController — listCourses
    √ returns 200 with courses array (3 ms)
    √ filters by type "Online"
    √ returns 500 on error
  courseController — createCourse
    √ returns 400 if title is missing (1 ms)
    √ returns 400 for invalid type (1 ms)
    √ returns 400 if Physical without location (1 ms)
    √ returns 400 if link is missing
    √ returns 400 if link is not a valid URL
    √ returns 201 with created course on valid input (1 ms)
    √ returns 201 for valid Physical course with location (1 ms)
    √ returns 500 on db error
  courseController — myCourses
    √ returns empty array if no courses found (1 ms)
    √ returns courses with application counts (1 ms)
    √ returns 500 on error
  courseController — updateCourse
    √ returns 404 if course not found (1 ms)
    √ returns 403 if user does not own the course
    √ returns 400 for invalid type during update
    √ returns 200 with updated course (1 ms)
  courseController — deleteCourse
    √ returns 404 if course not found
    √ returns 403 if not owner (1 ms)
    √ returns 200 after deleting course
    √ returns 500 on error (1 ms)

Test Suites: 1 passed, 1 total
Tests:       22 passed, 22 total
Snapshots:   0 total
Time:        0.534 s, estimated 1 s
Ran all test suites matching /courseController.unit/i.
```

2. Integration Testing
    1. Application & matching

```
PASS  src/tests/integration/application.integration.test.js
  Application API — Integration Tests
    POST /api/applications/apply/:internshipId
      √ 201 — submits a new application (34 ms)
      √ 201 — response contains application object (3 ms)
    GET /api/applications/my-applications
      √ 200 — returns list of my applications (3 ms)
      √ 200 — data array has correct number of items (3 ms)
    GET /api/applications/check/:internshipId
      √ 200 — returns application status check (3 ms)
      √ 200 — hasApplied is true when applied (3 ms)
    GET /api/applications/:id
      √ 200 — returns specific application by ID (3 ms)
      √ 404 — returns error for non-existent application (3 ms)
    PUT /api/applications/:id
      √ 200 — updates existing application (5 ms)
      √ 200 — update with email change (4 ms)
    DELETE /api/applications/:id
      √ 200 — withdraws application successfully (3 ms)
      √ 404 — returns error for non-existent application (3 ms)
      √ 403 — returns error when user not authorized (3 ms)
    GET /api/applications/internship/:internshipId
      √ 200 — returns applications for an internship (org role) (3 ms)
      √ 200 — data array has 2 applications (3 ms)

Test Suites: 1 passed, 1 total
Tests:       15 passed, 15 total
Snapshots:   0 total
Time:        0.855 s, estimated 2 s
Ran all test suites matching /application.integration/i.
```

2. Auth

```
PASS src/tests/integration/auth.integration.test.js
  Auth API — Integration Tests
    POST /api/auth/register
      √ 201 — registers a new user (29 ms)
      √ 201 — registers with role "organization" (4 ms)
      √ 201 — registers with default "youth" role when no role provided (3 ms)
    POST /api/auth/login
      √ 200 — logs in with valid credentials (3 ms)
      √ 200 — response contains user object (4 ms)
    GET /api/auth/me
      √ 200 — returns authenticated user profile (3 ms)
      √ 200 — works without Authorization header (middleware is mocked) (3 ms)
    PUT /api/auth/update
      √ 200 — updates user email (3 ms)
      √ 200 — updates user password (4 ms)
    DELETE /api/auth/:id
      √ 200 — deletes user account (3 ms)
    Route Error Handling
      √ 404 — unknown route returns 404 (3 ms)
      √ 405 — wrong method on register returns 404 (Express default) (2 ms)

Test Suites: 1 passed, 1 total
Tests:       12 passed, 12 total
Snapshots:   0 total
Time:        0.754 s, estimated 2 s
Ran all test suites matching /auth.integration/i.
```

3. Profiles & Governance

```
PASS tests/integration/youthProfile.route.test.js
  YouthProfile API routes (integration)
    POST /api/profile
      √ 201 — youth creates profile (57 ms)
      √ 400 — duplicate profile for same user (23 ms)
rofile in list (19 ms)
    PUT /api/profile/:userId
      √ 200 — youth updates own profile (28 ms)
      √ 404 — update when own profile row missing (authorized but no document) (7 ms)
    POST /api/profile/:userId/upload-cv
      √ 200 — valid PDF metadata (18 ms)
      √ 400 — invalid file type (15 ms)
      √ 404 — upload when own profile row missing (8 ms)
    DELETE /api/profile/:userId
      √ 200 — youth deletes own profile (22 ms)

Test Suites: 1 passed, 1 total
Tests:       11 passed, 11 total
Snapshots:   0 total
Time:        1.882 s, estimated 3 s
Ran all test suites matching /tests\\integration\\youthProfile.route.test.js/i.
```

```
PASS tests/integration/orgProfile.route.test.js
  Organization profile API routes (integration)
    POST /api/organizations
      √ 201 — organization user creates profile (53 ms)
      √ 400 — duplicate profile for same user (21 ms)
      √ 403 — youth cannot create organization profile (10 ms)
    GET /api/organizations/:id
      √ 200 — admin sees all organization profiles (27 ms)
    PUT /api/organizations/:id
      √ 200 — owner updates organization (23 ms)
      √ 404 — admin updates non-existent id (9 ms)
    POST /api/organizations/:id/documents
      √ 200 — uploads valid document metadata (19 ms)
      √ 400 — invalid document metadata (18 ms)
      √ 404 — admin uploads to missing organization id (11 ms)
    DELETE /api/organizations/:id
      √ 200 — owner deletes organization profile (20 ms)

Test Suites: 1 passed, 1 total
Tests:       13 passed, 13 total
Snapshots:   0 total
Time:        2.254 s
Ran all test suites matching /tests\\integration\\orgProfile.route.test.js/i.
```

4. Internship

```
PASS src/routes/internshipRoute.test.js
  Internship API Integration Tests
    POST /api/internships
      √ should create a new internship and return 201 (27 ms)
      √ should accept Active status internship (5 ms)
    PUT /api/internships/:id
      √ should update internship and return 200 (3 ms)
    DELETE /api/internships/:id
      √ should delete internship and return 200 (3 ms)
    GET /api/internships/my-internships
      √ should return organizations internship list (4 ms)
    GET /api/internships/dashboard/stats
      √ should return dashboard statistics (4 ms)
    GET /api/internships/search
      √ should return search results (3 ms)
      √ should accept keyword query param (3 ms)
      √ should accept skills and location query params (4 ms)
    GET /api/internships/:id
      √ should return internship by id (3 ms)
    PUT /api/internships/view/:id
      √ should increment view count (3 ms)

Test Suites: 1 passed, 1 total
Tests:       11 passed, 11 total
Snapshots:   0 total
Time:        0.719 s, estimated 1 s
Ran all test suites matching /internshipRoute.test/i.
```

5. Course

```
PASS src/tests/integration/course.integration.test.js
  Course API — Integration Tests
    GET /api/courses
      √ 200 — returns all courses without auth (16 ms)
      √ 200 — filters by type=Online (7 ms)
      √ 200 — returns empty if no courses match filter (3 ms)
      √ 200 — search query param is accepted (3 ms)
    GET /api/courses/my
      √ 200 — returns courses with application counts (4 ms)
      √ 200 — two courses returned for org user (3 ms)
    POST /api/courses
      √ 201 — creates a new course (13 ms)
      √ 201 — response contains organizerId (3 ms)
      √ 400 — missing title (4 ms)
      √ 400 — invalid type (4 ms)
      √ 400 — missing link (2 ms)
    PUT /api/courses/:id
      √ 200 — updates an existing course (3 ms)
      √ 404 — course not found (3 ms)
      √ 403 — returns forbidden when not owner (3 ms)
    DELETE /api/courses/:id
      √ 200 — deletes a course (2 ms)
      √ 404 — course not found (2 ms)
      √ 403 — not owner returns forbidden (3 ms)

Test Suites: 1 passed, 1 total
Tests:       17 passed, 17 total
Snapshots:   0 total
Time:        0.786 s, estimated 2 s
Ran all test suites matching /course.integration/i.
```

2. Coverage Report

File	% Stmts	% Branch	% Funcs	% Lines	Uncovered Line #s
All files	31.61	26.91	33.53	32.07	
config	0	100	0	0	
cloudinary.js	0	100	100	0	4-6
seed.js	0	100	0	0	5-42
controllers	34.07	29.31	45.45	35.06	
advisorController.js	0	0	0	0	6-131
applicationController.js	34.05	26.08	50	35.87	11-30,42-127,134-187,255-256,289-290
authController.js	91.76	97.56	100	92.85	114-115,153-154,180-181
careerGuidanceController.js	0	0	0	0	6-180
courseController.js	84.69	71.76	100	85.55	27,35,136,139-145,151-158,167
internshipController.js	100	100	100	100	
organizationController.js	0	0	0	0	14-277
profileController.js	0	0	0	0	14-366
jobs	0	0	0	0	
weeklyCron.js	0	0	0	0	14-66
middleware	0	0	0	0	
auth.js	0	0	0	0	4-30
authMiddleware.js	0	0	0	0	4-45
chatThrottleMiddleware.js	0	0	0	0	5-34
fakeAuth.js	0	100	0	0	3-16
roleMiddleware.js	0	0	0	0	1-8
uploadCV.js	0	0	0	0	5-27
uploadMiddleware.js	0	100	100	0	3-5
models	27.27	100	0	27.27	
Application.js	0	100	0	0	3-90
OrganizationProfile.js	0	100	100	0	3-70
User.js	100	100	100	100	
YouthProfile.js	0	100	100	0	3-118
applicationModel.js	0	100	100	0	3-11
courseModel.js	0	100	100	0	3
internship.js	100	100	100	100	
routes	29.41	0	0	30.76	
applicationRoutes.js	100	100	100	100	
auth.js	100	100	100	100	
courseRoutes.js	100	100	100	100	
emailTestRoute.js	0	0	0	0	10-75
internshipRoute.js	0	100	100	0	17-67
organizationRoutes.js	0	100	100	0	12-55
profile.js	0	100	100	0	12-55
services	47.52	38.49	47.27	47.77	
internshipService.js	87.73	74.64	94.11	90.32	20,133-134,243-245,257-274
matchingService.js	60.34	50	58.82	62.26	10-11,32-37,72-142,205,218-230
organizationService.js	0	0	0	0	1-127
profileService.js	0	0	0	0	3-166
tests	0	0	0	0	
application.manual.js	0	0	0	0	4-34
scoreTest.js	0	100	0	0	2-104
utils	9.52	0	0	10	
emailService.js	26.47	0	0	28.12	16-20,74-85,94-112,124-139,151-168,187-202,219-236
geocode.js	10	0	0	10	4-26
openaiChatClient.js	0	0	0	0	4-66
openaiChatError.js	0	0	0	0	5-93
openaiConfig.js	0	0	0	0	6-17

```
est Suites: 2 failed, 8 passed, 10 total
ests:       7 failed, 172 passed, 179 total
napshots:   0 total
ime:        5.416 s
an all test suites.
```